

also been directly worked out by researchers [106]. In this chapter, we will initially review the dynamics of a rigid body, followed by an exploration of only the Euler-Lagrange method.

8.2 Review on Rigid Body Dynamics

In this section, fundamental definitions and relationships for the dynamic analysis of rigid bodies will be examined. Considering that in [Chapter 3](#) and in [Section 3.6](#) methods for determining the linear acceleration of a point and the angular acceleration of a body have been discussed, the subsequent focus will be on the mass properties of the rigid body, linear and angular momentum, and the kinetic energy of the rigid body.

8.2.1 Center of Mass

Engineering students become familiar with the concept of mass in basic physics courses. Mass can be referred to as a property of a body that causes its weight in a gravitational field. As illustrated in [Figure 8.1](#), the distribution of mass in a rigid body affects its size, shape, and weight. Considering the coordinate system $\{\mathcal{A}\}$, in which the mass properties of the body are measured, if we assume a uniform volumetric density ρ for the mass distribution within the body, the mass of a differential element of the body can be represented as ρdV , where dV is the infinitesimal volume of the element. If we denote the position vector of this element in the coordinate system $\{\mathcal{A}\}$ by $\mathbf{p}$, the total mass of the body is obtained by integrating the mass of its elements over the body's volume.

$$m = \int_V \rho dV \quad (8.1)$$

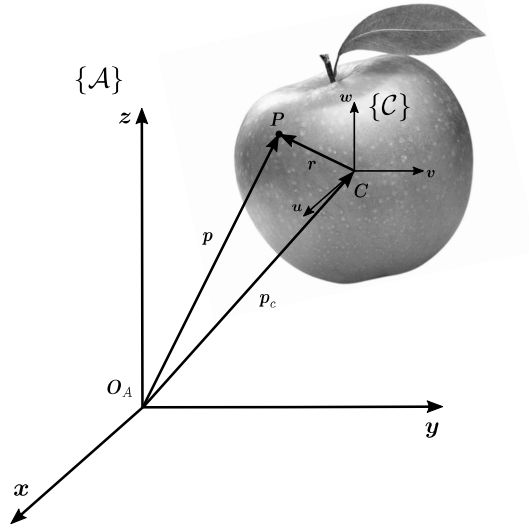
In that case, the position vector of the center of mass $\mathbf{C}$, denoted by $\mathbf{p}_c$ as depicted in the figure, can be determined based on the mass distribution within the body.

$$\mathbf{p}_c = \frac{1}{m} \int_V \mathbf{p} \rho dV \quad (8.2)$$

The center of mass is a theoretical point within an object where the entire mass of the object is concentrated, and it is evenly distributed around this point. Consequently, if the object is positioned on a sharp edge at its center of mass, the weight on both sides of the object is equal, leading to a state of equilibrium. In the dynamic analysis of rigid bodies, this point holds significant importance. Instead of scrutinizing the distributed mass within the body, one can simplify the dynamic equations by assuming the mass of the body is concentrated at this point. It is noteworthy that the mass properties of a rigid body are characterized by four parameters: the body's mass, denoted as m , and the three components of the position vector of its center of mass, $\mathbf{p}_c = [p_{cx} \quad p_{cy} \quad p_{cz}]^T$.

8.2.2 Moment of Inertia

Unlike the mass of a body that exhibits inertia against linear accelerations, the moment of inertia of a body manifests inertia against angular accelerations. In rotational motions, the mass distribution of a body with respect to the axis of rotation exhibits its moment of inertia. If we consider the rotational motion of a rigid body shown in [Figure 8.1](#), the moment of inertia of the body with respect to the reference coordinate system $\{\mathcal{A}\}$ is represented

**FIGURE 8.1**

Mass properties of a rigid body.

by the inertia matrix ${}^A\mathbf{I}$.

$${}^A\mathbf{I} = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix} \quad (8.3)$$

where:

$$\begin{aligned} I_{xx} &= \int_V (y^2 + z^2) \rho dV & I_{xy} &= I_{yx} = - \int_V xy \rho dV \\ I_{yy} &= \int_V (x^2 + z^2) \rho dV & I_{yz} &= I_{zy} = - \int_V yz \rho dV \\ I_{zz} &= \int_V (x^2 + y^2) \rho dV & I_{xz} &= I_{zx} = - \int_V xz \rho dV \end{aligned} \quad (8.4)$$

What is observed in these equations is that the diagonal components of the inertia matrix are directly proportionally to the square of the radial distance of the particle mass to the axis of rotation. This is in contrast to the non-diagonal components, which are proportionally to the product of the distance from the mass particle to the other axes. Thus, the inertia matrix of a body is a symmetric matrix, requiring only six dynamic parameters for its description. Additionally, as evident from the equations, the body's inertia is dependent on the coordinate system in which these calculations are performed. Typically, these calculations are done with respect to the coordinate system attached to the center of mass of the body $\{C\}$.

Calculating the inertia matrix for bodies is seldom done directly utilizing the definition provided in Equation 8.4. In contrast, for bodies with simple geometries, these computations are usually conducted using tables found in references [107, Appendix 5]. For bodies with intricate geometries, these computations are automated through computer-aided design (CAD) software. The only requisite step is to transform it from one coordinate system to another, a topic to be explored in the following sections.

8.2.3 Principal Axes

As mentioned, the inertia matrix of a body depends on the coordinate system with respect to which it is measured. For this reason, it is possible to find a coordinate system in space in which the inertia matrix becomes diagonal.

$${}^A\mathbf{I} = \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \quad (8.5)$$

As previously mentioned, coordinate axes in which the inertia matrix becomes diagonal are known as the principal axes of inertia. The diagonal components of this matrix in this configuration, denoted as I_{xx} , I_{yy} , I_{zz} are called the principal moments of inertia of the body. It can be demonstrated that the principal moments of inertia are among the fundamental mass properties of the body, easily obtainable through a transformation matrix consisting of the eigenvectors of the inertia matrix. Physically, the principal moments of inertia align with the symmetry axes of the body.

In this context, viewing the inertia matrix of a body as a mapping reveals that the principal components of the inertia matrix define the upper and lower bounds of this mapping. Furthermore, as the inertia matrix governs the mass properties of a body during rotational movements, the principal moments of inertia consistently remain positive. We will exploit these attributes when scrutinizing the characteristics of robot dynamic formulations.

8.2.4 Inertia Matrix Computations

As previously mentioned, the direct computation of body inertia matrices according to their definition is rarely performed. Typically, these values are derived from tables found in references or generated using CAD software. However, these inertia matrix values are commonly provided with respect to a coordinate system attached to the body at its center of mass or at its initial and final points. This stands in contrast to the dynamic analysis of bodies, where it becomes essential to ascertain inertia matrix values with respect to other coordinate systems. Consequently, comprehending the transformation of the inertia matrix from one coordinate system to another holds significant importance.

As illustrated in [Figure 8.1](#), if the change in coordinates in the inertia matrix is limited to translational motion, these calculations can be made based on the parallel-axis theorem [108] and using the following equation

$${}^A\mathbf{I} = {}^C\mathbf{I} + m(\mathbf{p}_c^T \mathbf{p}_c \mathbf{I}_{3 \times 3} - \mathbf{p}_c \mathbf{p}_c^T) \quad (8.6)$$

In this relation, ${}^A\mathbf{I}$ represents the inertia matrix relative to the coordinate system $\{\mathcal{A}\}$, and ${}^C\mathbf{I}$ is the inertia matrix relative to the coordinate system attached to the center of mass $\{\mathcal{C}\}$. Additionally, m denotes the mass of the body, the vector $\mathbf{p}_c = [x_c, y_c, z_c]^T$, as shown in [Figure 8.1](#), represents the translation vector from the coordinate system $\{\mathcal{C}\}$ to the coordinate system $\{\mathcal{A}\}$, and $\mathbf{I}_{3 \times 3}$ is the identity matrix of size 3×3 . This relation can be rewritten in terms of the components of the inertia matrix as follows.

$$\begin{aligned} {}^A I_{xx} &= {}^C I_{xx} + m(y_c^2 + z_c^2) & {}^A I_{xy} &= {}^C I_{xy} + m x_c y_c \\ {}^A I_{yy} &= {}^C I_{yy} + m(x_c^2 + z_c^2) & {}^A I_{yz} &= {}^C I_{yz} + m y_c z_c \\ {}^A I_{zz} &= {}^C I_{zz} + m(x_c^2 + y_c^2) & {}^A I_{xz} &= {}^C I_{xz} + m x_c z_c \end{aligned} \quad (8.7)$$

As observed in this relation, the diagonal components of the inertia matrix are increased proportional to the square of the radial distance between the two coordinate systems, and

the off-diagonal components are increased proportional to the product of the distances of the corresponding axes of the two coordinate systems.

If the coordinate transformation in determining the body's inertia matrix is limited to pure rotational motion, this transformation can be performed using the following equation.

$${}^A\mathbf{I} = {}^A\mathbf{R}_C {}^C\mathbf{I} {}^A\mathbf{R}_C^T \quad (8.8)$$

In this equation, ${}^A\mathbf{R}_C$ is the rotation matrix from the coordinate system $\{C\}$ to $\{A\}$. In the dynamic analysis of serial robots, the inertia matrix of the links is often obtained relative to the coordinate system located at their own center of mass. Equation 8.8 is then often used to transform it into other coordinate systems that have pure rotational motion relative to it.

8.2.5 Linear and Angular Momentum

The Newton-Euler and Euler-Lagrange methods are based on principles related to linear and angular momentum, as well as kinetic energy. A thorough exploration of this subject in diverse coordinate systems demands a more extensive investigation, which falls outside the scope of this book. Enthusiastic readers are encouraged to read the references provided in this area [107, 103]. In this context, we will concisely outline the outcomes of these studies, directly relevant to the dynamic analysis of serial robots.

The linear momentum of a rigid body can be obtained by multiplying its mass by the velocity of its center of mass.

$${}^A\mathbf{G} = m {}^A\mathbf{v}_c \quad (8.9)$$

In this context, it's important to emphasize that we consider the mass distribution around the center of mass as if it were concentrated solely at the center of mass itself. Consequently, we avoid computing individual particle momenta. This relationship is only valid using the velocity of the body's center of mass $\mathbf{v}_c$, and it cannot be determined using the velocities of other points on the body. Hence, the significance of precisely defining the center of mass and determining its velocity becomes evident in the dynamic analysis of rigid bodies.

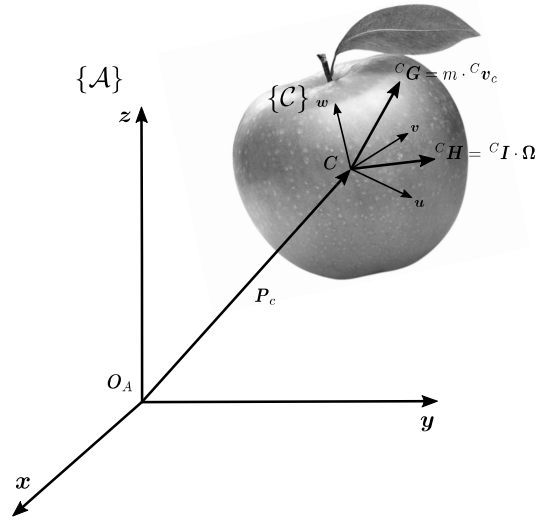
As indicated by Equation 8.9, the linear momentum of a body, akin to its center of mass velocity, is a vector with interpretability across various coordinate systems. In this equation, the linear momentum of the body with respect to the coordinate system $\{A\}$ is represented, and for this purpose, the velocity of the center of mass calculated in the coordinate system $\{A\}$ is used. Considering the vector properties of linear momentum, the homogeneous transformation matrix can be used to transform it into another coordinate system as follows.

$${}^A\mathbf{G} = {}^A\mathbf{T}_C {}^C\mathbf{G} = m {}^A\mathbf{T}_C {}^C\mathbf{v}_c \quad (8.10)$$

The angular momentum of a rigid body is also a vector, calculated with respect to the coordinate system attached to the body at its center of mass $\{C\}$ as follows.

$${}^C\mathbf{H} = {}^C\mathbf{I}\boldsymbol{\Omega} \quad (8.11)$$

With this definition, angular momentum extends the concept of linear momentum to rotational motions. Given that the angular velocity of the body $\boldsymbol{\Omega}$ is the same for all points on the body, calculating angular momentum only requires determining the inertia matrix of the body in the $\{C\}$ coordinate system.

**FIGURE 8.2**

Linear and angular momentum of a rigid body about the $\{C\}$ reference frame.

If we intend to calculate angular momentum with respect to another coordinate system, such as $\{A\}$, it is no longer straightforward to accomplish this using a homogeneous transformation matrix. Instead, we need to use the following equation:

$${}^A\mathbf{H} = \mathbf{p}_c \times \mathbf{G}_c + {}^C\mathbf{H} \quad (8.12)$$

As observed in this equation, to determine the angular momentum of the body with respect to the coordinate system $\{A\}$, denoted by ${}^A\mathbf{H}$, it is necessary to sum the angular momentum of the body with respect to the center of mass coordinate system $\{C\}$ with the cross product of the vector representing the distance from the center of mass to the origin of the coordinate system $\mathbf{p}_c$ and the linear momentum $\mathbf{G}_c$.

Note that in this equation, the mass distribution around the center of mass is also treated as a concentrated mass at the center of mass. Consequently, for calculating the angular momentum with respect to the coordinate system $\{A\}$, it suffices to combine the angular momentum with respect to the center of mass coordinate system $\{C\}$ and the effect of the linear momentum of the body about the $\{A\}$ coordinate system.

8.2.6 Kinetic Energy

In the analysis of dynamics using the Euler-Lagrange method, the kinetic energy of a rigid body plays a crucial role. To calculate the kinetic energy of a rigid body, it is sufficient to obtain the total kinetic energy of the linear and angular motion of the body with respect to the coordinate system attached to the system's center of mass.

$$K = \frac{1}{2} \mathbf{v}_c \cdot {}^C\mathbf{G} + \frac{1}{2} \boldsymbol{\Omega} \cdot {}^C\mathbf{H} \quad (8.13)$$

In this equation, the kinetic energy of the body is a scalar quantity. Its linear motion component is calculated by the dot product of the linear velocity of the center of mass and its linear momentum, while the angular motion component is determined by the dot product of the angular velocity of the body and its angular momentum about the center

of mass. If the motion of the body is confined to the $x - y$ plane, the following simplified relationship can be utilized.

$$K = \frac{1}{2}mv_c^2 + \frac{1}{2}{}^C I_{zz}\Omega_z^2 \quad (8.14)$$

In this equation, $\mathbf{v}_c = [v_{c_x}, v_{c_y}]^T$ represents the velocity of the center of mass of the body in the $x - y$ plane, and Ω_z is the angular velocity of the body about the z axis. It is for this reason that, in determining the angular momentum, only the inertia component of the body with respect to the coordinate system centered at the center of mass and about the z axis, denoted by ${}^C I_{zz}$, has been used. This equation will be employed in dynamic formulation for planar robots.

8.3 Euler-Lagrange Method

The Euler-Lagrange method is derived by rewriting classical mechanics, combining the principles of conservation of linear and angular momentum with energy conservation. This method is particularly effective for dynamic analysis of multibody systems, offering efficiency by eliminating the need to account for support forces between system components in the equations of motion. As a result, the analysis is considerably simplified.

In the Euler-Lagrange method, the equations of motion for a system are determined using a set of *generalized coordinates*. These coordinates represent the minimal set of variables necessary to describe the system's motion, with the understanding that they are not uniquely defined. Consider the motion of a bead in a narrow groove, for example. If we attempt to derive the equations of motion for this system using the Newton-Euler method, we would need to determine the friction force between the bead and the groove. This force is often nonlinear and time-varying, making its determination challenging. However, without the need to consider this force and by solely considering the path of the bead's motion through generalized coordinates, we can easily derive its equations of motion. With this choice and without calculating support forces between different bodies, the system's dynamics can be analyzed.

By defining generalized coordinates, external forces and torques acting along these coordinates are determined as *generalized forces*. Subsequently, the kinetic and potential energy of the moving masses in the system are calculated in terms of generalized coordinates. Finally, the Euler-Lagrange equations are employed to determine the equations of motion. The details of this method will be thoroughly discussed in the following sections.

8.3.1 Generalized Coordinates and Forces

The connection between two masses in multi-body systems is described by kinematic constraints that restrict the relative motion between them in various directions. In serial robots, joints create these kinematic constraints, defining and limiting the system's degrees of freedom.

In a general classification, system constraints are termed "holonomic" if the relationship between position variables can be expressed by an algebraic equation between these variables and, if necessary, the time.

$$f(x_1, x_2, \dots, t) = 0 \quad (8.15)$$

In this algebraic relationship, x_i represents position or orientation variables, while their derivatives are not seen in this equation. For example, consider the angular motion of a